

horizontal stabilizer or *tailplane*, and a rotating aft section, called the elevator. Both the horizontal stabilizer and the elevator contribute to pitch stability, but only the elevator provides pitch control. In some cases the entire horizontal tail rotates to provide pitch control. This is called a *simple stabilator* or an *all-flying tail*. On some airplanes both the front stabilizer section and the aft elevator section of the horizontal tail rotate separately to provide pitch control. This configuration is called a *compound stabilator*. The compound stabilator looks and responds very much like the conventional horizontal tail, but it differs somewhat in its mechanical operation.



Figure 4.3.1. British Spitfire, showing the conventional aft tail. (Photograph by Barry Santana)

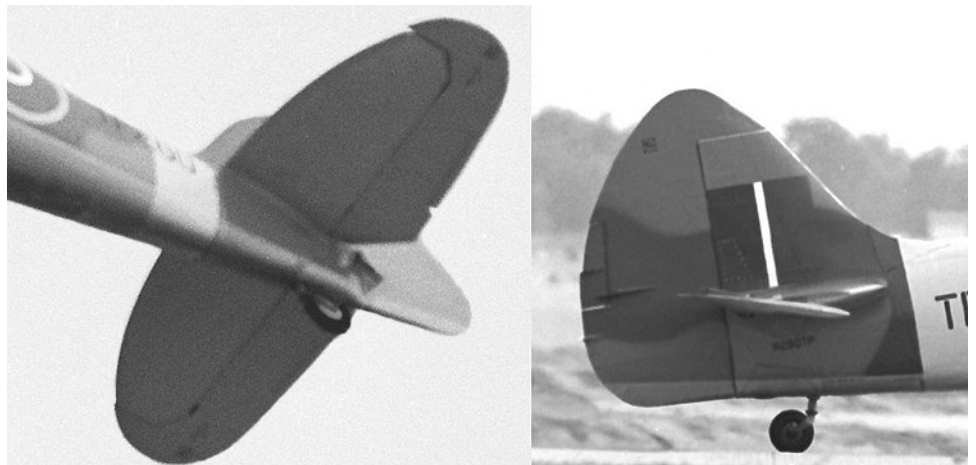


Figure 4.3.2. Empennage of the British Spitfire. (Photographs by Barry Santana)